

World as a Puzzling System: An Essay

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Abstract

Causal reasoning is often pictured as an arrow: X happens, and Y follows. That picture is useful, but the world rarely stops at Y. Consequences alter later conditions; policies reshape politics; behavior reinforces or weakens norms; people act on explanations of what has happened; and those actions become part of what happens next. This essay examines what causal attribution looks like when the object of study is not an isolated cause–effect pair but an evolving system. It begins with a simple puzzle—do the times make the hero, or does the hero make the times?—and follows the puzzle through temporal ordering, feedback, stable states, trajectories, intervention, and adaptation. A central distinction is between a causal mechanism and our ability to distinguish it from competing explanations. The essay also considers why the demand for causal explanation can become especially strong when the evidence remains ambiguous, and why acting on an attribution can change the system to which the attribution referred. The aim is not to replace ordinary cause-and-effect reasoning, but to place it inside a world that continues moving while we try to understand it.

1. Do the Times Make the Hero?

An old question can be stated in two opposite ways: Do the times make the hero, or does the hero make the times?

Both statements are easy to understand. Both can also sound persuasive. Research on historical institutional change has long emphasized that some periods loosen established constraints and make actors' choices unusually consequential, while institutions themselves are also created, maintained, and changed through action [1]. Yet once a leader becomes consequential, that person can redirect events, alter institutions, change expectations, and leave the world different from the one that first made the leadership possible.

The puzzle is not solved by declaring that both statements are true. That answer is too easy. The more interesting question is why two causal attributions pointing in opposite directions can remain plausible when they refer to the same historical development.

Imagine a highly simplified sequence. A changing social environment S_0 creates conditions under which an actor H_1 becomes consequential. The actor then changes the environment, producing S_2 . Those changes alter what the actor can do next, producing H_3 , and the sequence continues:

$$S_0 \rightarrow H_1 \rightarrow S_2 \rightarrow H_3 \rightarrow \cdots$$

An observer concentrating on the first part of the sequence may reasonably emphasize the times making the hero. Another concentrating on the next part may emphasize the hero making the times. Their disagreement may reflect different causal claims, but it may also reflect different slices of the same trajectory.

This pattern is hardly confined to historical leaders. Do institutions shape behavior, or does repeated behavior reproduce institutions? Does technology transform society, or do social needs and institutions shape

technology? Does politics make policy, or does policy reshape politics? The questions recur because the objects being compared do not stand still while we study them.

The persistence of such debates gives us a modest but important starting point: attribution is hard. The difficulty need not mean that causes are unknowable, or that every explanation is equally good. It means that the familiar task of assigning an outcome to a cause becomes more demanding when causes and consequences continue interacting after the moment we choose to examine.

To see why, it helps to begin with a much easier world.

2. The Comfortable World of $X \rightarrow Y$

Suppose a switch is flipped and a lamp turns on. We can complicate even this example if we wish—the electrical system may fail, the bulb may be defective, or someone may have changed the wiring—but the basic causal picture is comfortable. An event X occurs, and an event Y follows.

$$X(t) \rightarrow Y(t + \Delta t), \quad \Delta t > 0$$

Time gives the two events an ordering. X occurs first; Y occurs later. Temporal precedence does not prove that X caused Y . A common cause may have produced both, the sequence may be coincidental, or important variables may be missing. But the ordering gives causal reasoning a useful structure: the candidate cause is earlier than the candidate consequence.

Prediction and attribution then appear naturally different. Standing at X , we can ask what Y will follow. Standing at Y , we can ask what earlier X helped produce it. Prediction looks forward along the temporal sequence; attribution looks backward.

This simple picture is extraordinarily useful. Much of everyday reasoning, scientific experimentation, policy analysis, and management depends on finding relationships that can be represented well enough in this form. Nothing in this essay requires abandoning it.

The difficulty begins when we quietly assume that the arrow ends at Y .

3. Let the World Keep Moving

Consider public policy. Politics helps determine which policies are enacted. Elections, coalitions, public opinion, institutions, interest groups, and political leadership all influence policy outcomes. Written as a simple causal story, politics produces policy.

$$\text{politics} \rightarrow \text{policy}$$

But enacted policy does not disappear after becoming an outcome. It changes the world in which subsequent politics occurs. A policy can distribute resources, create beneficiaries, alter administrative capacities, change what citizens expect from government, affect participation, and reshape later political preferences. Policy-feedback research has documented precisely this reversal: policies are not only products of politics; they can become inputs into later politics [2,3].

$$\text{politics}_t \rightarrow \text{policy}_{t+1} \rightarrow \text{politics}_{t+2}$$

Nothing mysterious has happened. We have simply allowed time to continue.

Yet that small move changes the way we should talk about cause and effect. Policy was an effect in the first relationship and a cause in the second. Politics was a cause at one point and a consequence later. The labels have not become meaningless; they have become indexed to a particular part of the trajectory.

The same structure appears elsewhere. A norm influences behavior, while repeated behavior reinforces or weakens the norm. Market conditions shape firms, while firms alter markets. A social movement elevates leaders, while leaders redirect the movement. Technology changes organizations, while organizational use changes which technologies are developed next.

The general pattern is feedback:

$$X_t \rightarrow Y_{t+1} \rightarrow X_{t+2} \rightarrow Y_{t+3} \rightarrow \dots$$

Once feedback enters the picture, a variable need not carry a permanent identity as either cause or effect. Its role depends on where we stand along the path. The current state can also contain traces of what happened earlier: today's political preferences may partly reflect yesterday's policies; today's institutions may embody earlier responses to problems; today's behavior may reflect expectations formed by previous behavior.

This is not a new observation. Dynamic and cyclic causal models explicitly allow systems whose components evolve through feedback, and equilibrium causal models show that long-run causal relations can sometimes be studied even when the underlying system contains cycles [4]. The point here is simpler: established ideas about feedback immediately complicate the intuitive act of looking at two observed quantities and asking which one made the other.

That complication also changes the opening puzzle. The times may help make the hero at one point along the trajectory, and the hero may help make later times at another. Whether either attribution is adequate depends on what question we are asking, what part of the path we observe, and what competing explanations the evidence can rule out.

The next difficulty appears when this continuous activity produces very little visible change. A system can keep working, responding, and feeding back while looking remarkably stable.

4. A Stable World Can Be Very Busy

Feedback does not always produce visible movement. Sometimes its most important effect is to keep a system where it already is.

Consider a social norm. A norm influences behavior because people expect others to follow it, anticipate approval or disapproval, or simply regard the behavior as normal. Research on norm dynamics emphasizes precisely these expectation-based and self-reinforcing processes, as well as the ways norms persist and change over time [5,6]. Every act of compliance also contributes to the environment in which the next person forms those expectations. The norm helps produce the behavior, and the behavior helps reproduce the norm.

$$\text{norm} \rightarrow \text{behavior} \rightarrow \text{expectations} \rightarrow \text{norm}'$$

If this loop persists, the system can look almost motionless from a distance. The same norm is present this year as last year; the same broad pattern of behavior continues. Yet the apparent stability is not evidence that nothing is happening. People are continuously acting, observing one another, forming expectations,

complying, resisting, and adjusting. Activity can be substantial even when the higher-level state changes very little.

This distinction matters because stability is easy to mistake for causal simplicity. A maintained state can be the visible result of continuous feedback underneath it.

There is another way stability can mislead. Imagine two systems that show almost no net movement. In the first, little is pushing the system in either direction. In the second, powerful opposing forces nearly cancel. The observed stability may look similar, but the systems are not similar at all. A small disturbance might do almost nothing to the first and tip the second into a very different state.

A closely divided political system offers an intuitive example. In the United States, overall partisan alignment has remained near balance even as the two parties and their supporters have become sharply differentiated on many dimensions [7,8]. Aggregate stability need not imply internal harmony.

So a stable state tells us something important: where the system is. It does not necessarily tell us what keeps the system there.

That raises a more precise causal question. If two different underlying mechanisms can sustain the same observed state, can the state itself tell us which mechanism is operating?

5. What the State Does Not Tell Us

A small thought experiment makes the problem concrete.

Suppose two systems contain the same variables, X and Y , and both eventually settle at the same observed state:

$$(X^*, Y^*) = (0, 0)$$

In the first system, X affects Y . In the second, Y affects X . If we observe only the settled state, both systems give us exactly the same picture: X is zero and Y is zero. The causal mechanisms are different, but the observation does not distinguish them. This illustrates the general problem: identical observed states can arise from distinct causal architectures.

Now disturb X slightly.

If X affects Y , the disturbance propagates to Y . If the true direction runs from Y to X , the same disturbance need not produce that response in Y . The mechanisms that looked identical at the maintained state begin to separate along the response path.

The example is deliberately simple. Its purpose is not to show that equilibrium states are generally uninformative. They are not. Equilibrium causal models show that, under suitable assumptions and designs, equilibrium or cross-sectional observations can support causal inference about long-run interventions [4]. The narrower point is that a causal mechanism and the evidence available to distinguish that mechanism are different things.

This motivates the first of two analytical constructs used in the essay: causal separability.

Causal separability (CS) is how well the available evidence distinguishes competing causal explanations.

This is not the same as asking whether a causal relationship exists or how strong its effect is. Two mechanisms can be genuinely different while a particular set of observations leaves them indistinguishable.

The distinction is useful because causal arguments often move too quickly from an observed state to a causal story. If X and Y coexist in a stable relationship, we may be tempted to narrate X as the cause of Y, Y as the cause of X, or some third factor as the cause of both. But the observed state may not contain enough information to choose among those stories.

What additional evidence would help? One possibility is to stop looking only at the state and watch what happens when the system moves.

6. The Path Can Tell a Different Story

Return to the two systems, but make the problem harder. Suppose both contain genuine feedback between X and Y. They settle at the same state. We perturb X, and even their first responses are identical. At that moment, the observation still cannot tell us which feedback mechanism is operating.

Then the feedback has time to travel through the system.

X changes Y. Y feeds back into X. If the two systems differ in the strength or structure of that return path, their later states begin to diverge. Eventually both may settle back to exactly the same equilibrium.

same equilibrium → same first response → different paths → same equilibrium

The causal difference exists in the middle.

If we record only the state before the disturbance and the state after the system has settled again, we can miss it completely. If we observe the trajectory, the difference becomes visible.

This gives us three observational objects that are easy to blur together: a state at one moment; the difference between two endpoints; and the trajectory connecting them. They need not contain the same causal information.

The time horizon matters as well. Observe too soon and a feedback effect may not yet have propagated far enough to distinguish competing mechanisms. Observe the adjustment path and the difference may become clear. Observe only the eventual settled state and the evidence may again appear to disappear. Temporal causal-discovery research likewise treats sampling frequency and subsampling as consequential for what causal structure can be recovered [9]. The opening hero-and-times puzzle has the same temporal vulnerability: a short slice can emphasize the times shaping the actor, while a later slice emphasizes the actor reshaping the times.

None of this means that trajectories automatically reveal causality. A hidden common driver can produce a persuasive temporal pattern without X causing Y at all. Perfect-adaptation models provide a particularly useful established example: a variable can respond transiently to a persistent intervention and later return to its original equilibrium value, so equilibrium observations can conceal features of the causal response path [10]. The point is narrower and more useful: paths can contain discriminating information that endpoints do not.

This is why perturbation is so powerful in science. Rather than merely observing where a system sits, we disturb something and watch how the system responds.

That seems to offer an obvious solution to the attribution problem: if we cannot tell what causes what, intervene.

But in a social system, intervention creates a new problem. The system may respond not only by revealing how it works, but by changing how it works.

7. The Intervention That Changes the Answer

Attribution matters because people act on it.

Suppose we observe a relationship between X and Y , conclude that changing X will improve Y , and design an intervention accordingly. In the simplest case, the underlying relationship remains stable. We change X and learn whether Y responds as expected.

$$Y = f(X)$$

Many social systems are less cooperative. People observe the intervention. They change expectations, strategies, timing, and behavior. Organizations redesign processes. Markets reprice. Political coalitions form around the new rule. Institutions adapt. The intervention that entered our analysis as something imposed from outside becomes part of the environment to which the system responds.

The relevant relationship may therefore change with the intervention: The important change is not merely in the values of X and Y ; the function relating them can change.

$$Y_t = f_t(X_t) \rightarrow Y_{t+1} = f_{t+1}(X_{t+1})$$

An intervention can be exogenous at entry and become endogenous through response.

This possibility is familiar in economics. The Lucas critique, for example, warned against evaluating policy changes using behavioral relationships estimated under a policy regime that the proposed intervention itself would alter. If people change their decision rules when policy rules change, a historical relationship need not remain invariant after intervention [11].

The broader idea extends beyond macroeconomic policy. A regulation can induce avoidance strategies. A performance target can change the behavior being measured. A subsidy can change prices and entry decisions. A successful safeguard can reduce the very problem that originally made the safeguard appear necessary. These mechanisms are not identical: some interventions change behavior or decision rules, while successful prevention can instead reduce the problem that called for the intervention. In each case, intervention does more than move one variable while the rest of the system waits.

This produces an uncomfortable tension: intervention can reveal structure while also altering structure. It is one of our strongest tools for learning causal relationships, yet in an adaptive system the intervention that reveals a relationship can also help make that relationship obsolete.

The point is not that intervention is futile. Interventions can work extraordinarily well. The point is that their effects belong to a trajectory. The system after intervention is not necessarily the same system with one variable changed.

And there is another complication. We have so far spoken as if an observer first understands the system and then intervenes. In many social settings, observers are already inside the process. Their expectations, predictions, and causal stories affect what they do—and what they do affects what happens next.

The current AI boom gives us a living example.

8. The Observer Enters the System

The current AI boom gives us a useful case because its endpoint is unknown.

When we study the New Deal, the rise of a historical leader, or the spread of an earlier technology, we know much of what came afterward. That knowledge is helpful, but it also makes retrospective stories deceptively easy to construct. Once an outcome is known, a long and uncertain path can be compressed into a clean sequence of causes that appears more obvious than it did while events were unfolding.

AI presents a different problem. We are inside the trajectory.

$$S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow ?$$

Enormous investment, data-center construction, organizational adoption, changing work practices, and expectations about future productivity are all visible. Wachter and Wachter report that the five largest U.S. technology firms spent \$380 billion on capital expenditure in 2025 and were forecast to spend roughly twice that amount in 2026 [12]. Nationally representative U.S. Census data also show AI spreading through firms, business functions, and worker tasks, although adoption remains uneven and often limited in scope [13]. What these developments will ultimately amount to is not.

One causal story begins with capability. Better AI makes valuable applications possible; firms invest; adoption spreads; productivity eventually rises. Recent executive surveys report positive but heterogeneous productivity effects and expectations of stronger gains, while broader reviews describe the empirical productivity evidence as real but uneven and still difficult to translate into aggregate outcomes [14,15].

$$\text{capability} \rightarrow \text{investment} \rightarrow \text{adoption} \rightarrow \text{productivity}$$

Another begins with expectations. Belief in transformative future value attracts capital; capital finances infrastructure and model development; rising valuations reinforce expectations; those expectations induce still more investment.

$$\text{expectations} \rightarrow \text{investment} \rightarrow \text{valuations} \rightarrow \text{expectations'}$$

A third story emphasizes the long adjustment path. AI requires infrastructure, organizational redesign, new skills, complementary technologies, and experimentation before broad productivity effects become visible. Current evidence already shows substantial organizational and task-level adaptation, while energy agencies document a major data-center buildout whose scale and payoff depend on uncertain AI uptake, efficiency, financing, and infrastructure constraints [13,15,16].

$$\text{AI} \rightarrow \text{infrastructure} \rightarrow \text{organizational redesign} \rightarrow \text{delayed payoff}$$

These stories are not mutually exclusive. That is precisely the difficulty. The same current observations can fit several accounts, and the accounts may become empirically more separable only as the trajectory unfolds.

The problem is therefore different from retrospective attribution. Historical analysis risks trajectory compression: once we know where the path ended, we make the route look cleaner than it was. A live case suffers trajectory incompleteness: we do not yet possess the later observations that may distinguish competing explanations.

AI adds another complication. The explanations themselves affect the trajectory. Expectations about productivity can influence investment; forecasts about labor displacement can affect workers, firms, educators, and policymakers; and expectations about electricity demand and returns influence infrastructure decisions [12,16,17]. This possibility has a long social-science lineage: reflexive or self-altering predictions can change behavior in ways that help produce or defeat the predicted outcome [18]. The observer is not entirely outside the system.

system → observation → prediction / attribution → action → system'

A prediction can therefore become one of the causes of the future against which the prediction will later be judged. An attribution can encourage investment, regulation, resistance, or imitation and thereby alter the phenomenon being attributed.

This does not make the AI boom unknowable. It makes it an unusually clear example of causal reasoning conducted while the causal system is still responding to the reasoning.

And it raises a further question. AI attracts extraordinary demand for explanation. But does intense demand for a causal story mean that the available evidence can actually distinguish among the stories?

9. When the Demand for Explanation Outruns the Evidence

Imagine a political system divided almost exactly in half:

49.9% versus 50.1%

The numbers are stylized, but the structure is familiar. U.S. registered voters were nearly evenly divided in partisan alignment in 2024, and the 2024 presidential popular vote was decided by 1.5 percentage points [7,19]. At the same time, party loyalty in vote choice remained extremely high, and research documents substantial affective polarization between partisans [8,20]. Two large coalitions can therefore remain close to numerical balance while differing sharply in identities, priorities, and political attachments.

Is such a system stable or unstable?

At one level it can be remarkably stable. The underlying division may remain near half-and-half across repeated contests. At another level it can be highly volatile. A movement of only a fraction of the electorate can cross an institutional threshold and transfer control from one coalition to the other.

small change in underlying state → large change in institutional outcome

The magnitude of system change is therefore not the same thing as the magnitude of consequence.

Nor does aggregate stability imply internal harmony. A nearly stationary balance can result from two weak forces, or from two strong forces pushing in opposite directions. The visible aggregate alone does not tell us which system we inhabit.

Highly competitive politics also generates an enormous appetite for causal explanation. Why did one side win? Was it the economy, turnout, candidate quality, policy, demographic change, media, a particular event, or reaction against the incumbent? When margins are small, many factors can plausibly exceed the margin itself. The demand to identify the decisive cause can be intense even when causal attribution is unusually difficult.

The political example motivates the second analytical construct used in the essay: causal salience. Where causal separability concerns the evidence available to distinguish competing explanations, causal salience concerns the pressure to ask the causal question in the first place.

Causal salience (CA) is how strongly a causal question attracts attention, demands explanation, or motivates intervention.

The distinction is not about whether a cause is objectively important. CA is close to ordinary issue salience, but narrower: it refers specifically to salience routed through a causal question. A relationship can be highly salient because observers perceive its stakes as high even when the available evidence does not distinguish well among competing explanations.

Taken together, CS and CA describe two different dimensions of an attribution problem:

Four combinations are possible: low CA / low CS (little pressure, weak discrimination); low CA / high CS (stronger causal discrimination but little pressure to explain); high CA / low CS (strong pressure, weak discrimination); and high CA / high CS (strong pressure with relatively discriminating evidence).

The high-CA/low-CS case is especially consequential: pressure to explain and act is strong while causal discrimination remains weak. A crisis, close election, market disruption, or technological boom can occupy this region.

CA ≠ CS

The two can move together, but they need not. A clean experiment can have high causal separability while attracting little public attention. A crisis, election, market crash, or technological boom can have enormous causal salience while supporting several competing explanations. A market crash, for example, can generate an urgent search for a single cause while several shocks, feedback loops, and expectations moved together.

The AI boom illustrates the same divergence in a different setting. Questions about productivity, employment, valuation, investment, energy, and organizational change are highly salient. The evidence is also mixed in ways that matter for attribution: a 2026 ILO review finds real but uneven productivity gains and limited large-scale displacement so far, while recent U.S. payroll evidence finds no economy-wide displacement but a widening employment gap for young workers in highly AI-exposed occupations [15,17]. Many causal claims will become easier—or harder—to distinguish only as the trajectory continues.

This matters because salience encourages action. When CA is high and CS is low, people may invest, regulate, vote, reorganize, imitate, or resist under strong pressure to act but limited ability to distinguish among causal stories. Those actions then enter the next state of the system.

At this point the original question—does X cause Y or does Y cause X?—can begin to feel too narrow. Sometimes it remains exactly the right question. Sometimes, however, the more revealing object is the evolving relationship between them.

10. Which Made Which?

Return to the question that opened the essay: Do the times make the hero, or does the hero make the times?

We can now see several reasons the debate persists. One observer may focus on the conditions under which an actor becomes consequential; another on what the actor subsequently changes. They may observe different portions of a feedback path. They may work at different levels—individual and society—or over different time horizons. And the relationship itself may evolve as the interaction proceeds.

The same structure appears in other familiar debates. Does technology shape society, or does society shape technology? Do institutions shape behavior, or does behavior reproduce and change institutions? Does a movement produce its leaders, or do leaders produce the movement?

Feedback alone does not make directional causality meaningless. At a particular point we may still identify a clear effect of X on Y. The stronger complication appears when interaction changes not only X and Y but also how they will affect one another next time. The key part in the representation below is the time subscript on the functions themselves: not only the variables, but the mappings between them may evolve.

$$X_{t+1} = f_t(X_t, Y_t) \quad Y_{t+1} = g_t(X_t, Y_t)$$

Suppose a new technology changes an organization. The organization redesigns its workflow around the technology. That redesign changes what users need, which changes what developers build next. The technology at the next stage is not acting on the same organization, and the organization is not responding to the same technology.

Research on the social construction of technology has long challenged simple technological determinism by examining how technological artifacts and relevant social groups shape one another [21]. Coevolutionary research makes a related point more generally: technologies, institutions, organizations, and ideas can change through mutually conditioning processes rather than one merely acting on a passive other [22].

In such settings, a fixed arrow can remain useful locally. It may be perfectly meaningful to ask whether a particular technology changed a particular work practice. But the arrow can become an incomplete summary of the longer trajectory.

This is not a reason to replace every causal question with the vague claim that everything affects everything. Quite the opposite. The challenge is to identify where directional claims remain separable and where the evolving interaction itself becomes part of the explanation.

Sometimes an observed pattern can also tell us something about the larger system in which those local causal relationships occur.

11. What Does the Pattern Reveal?

If an evolving relationship cannot always be summarized by one fixed arrow, the next question is what the broader pattern can tell us about the system in which the arrows operate. Scientific discovery offers a useful example. Newton, Darwin, and many others are remembered because particular ideas became inseparable from particular names. The natural causal story is individual-centered: exceptional ability produces exceptional discovery.

Yet the history of science repeatedly contains cases of multiple or near-simultaneous discovery. Robert Merton's classic studies treated such 'multiples' as a recurring feature of scientific history [23,24]. Such cases do not make individual ability irrelevant, nor do they prove that a discovery was inevitable. They do suggest that something about the state of a field matters.

Merton specifically emphasized the accumulation of prerequisite knowledge and tools, focused attention, and scientific or social needs as conditions under which similar discoveries may arise independently [24]. When several independent investigators approach the same discovery, the pattern itself becomes evidence about the environment in which those investigators were working.

$$\text{individual capability} + \text{field conditions} + \text{trajectory} \rightarrow \text{realized discovery}$$

The question has shifted slightly. We can still ask who caused what. We can still compare rival causal explanations. But we can also ask what the observed pattern reveals about the larger system.

This form of reasoning is common in science. An observable phenomenon can matter not only because we want to explain that phenomenon, but because it contains information about a process or state that is not directly visible. The inference is only as good as the model connecting observation to system, but the direction of inquiry is important.

For an evolving social system, the same possibility arises. A cluster of innovations, a burst of legislation, repeated institutional conflict, or the emergence of several similar organizational responses may tell us something about the system conditions under which those events became possible.

This does not eliminate local causal attribution. It places local attribution alongside another question: what kind of system produces the pattern we are observing?

That question brings us back to the world with which we began.

12. A Puzzling System

Do the times make the hero, or does the hero make the times?

We do not need a slogan to resolve the question.

Sometimes one direction can be identified with convincing evidence. Sometimes the times create an opportunity and the actor subsequently changes the times. Sometimes a feedback loop makes each direction relevant at a different point. Sometimes the answer depends on the time horizon or level at which we ask the question. Sometimes an intervention helps separate competing explanations. Sometimes the intervention changes the system we were trying to understand.

The simple causal arrow remains indispensable. The central lesson is not to abandon it, but to follow the system beyond it.

$$X \rightarrow Y$$

But an evolving system asks us to keep watching:

$$X \rightarrow Y \rightarrow X' \rightarrow Y' \rightarrow \dots$$

Once the path continues, effects can become causes. Stable states can be maintained by active feedback. Different mechanisms can produce the same observed state. Trajectories can reveal distinctions that

endpoints conceal. Observers can act on their attributions, and their actions can change subsequent behavior and sometimes the mechanism itself.

This is why attribution can be difficult even when causes are real. The difficulty is not always a defect in reasoning or a shortage of data. Sometimes it arises because the object of attribution continues to evolve while we observe it.

That does not excuse careless causal claims. It makes disciplined causal reasoning more important. We still need evidence capable of distinguishing explanations, especially when causal salience is high and pressure to intervene is strong. We still need interventions, comparisons, counterfactual reasoning, historical analysis, and models whose assumptions can survive scrutiny. But we should also ask whether the causal question has been framed at the right point along the trajectory and at the right level of the system.

The world may be puzzling not because it lacks causes, but because causes do not wait patiently for us to finish attributing them before becoming consequences, feedback, beliefs, interventions, and causes again.

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